

**INDIRA GANDHI DELHI TECHNICAL UNIVERSITY FOR
WOMEN**

KASHMERE GATE, DELHI-110006



(BEC-205) ANALOG AND DIGITAL ELECTRONICS LAB

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INDEX

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EXPERIMENT-1

AIM: To study and design basic electronic circuits using **LTspice** simulation software.

APPARATUS: LTspice software

THEORY: LTspice is a free, high-performance SPICE-based analog electronic circuit simulator developed by Analog Devices. It allows students and engineers to design, simulate, and analyse electronic circuits without physically assembling them.

Key features:

1. Schematic Capture – Easy drawing of circuits using a graphical interface.
2. Simulation – Performs transient analysis, DC operating point analysis, AC analysis, and Fourier analysis.
3. Library Support – Includes models of resistors, capacitors, diodes, BJTs, MOSFETs, op-amps, and ICs.
4. Waveform Viewer – Displays voltages, currents, and power in graphical form.
5. Parameter Sweeps – Allows variation of component values to study circuit behaviour.

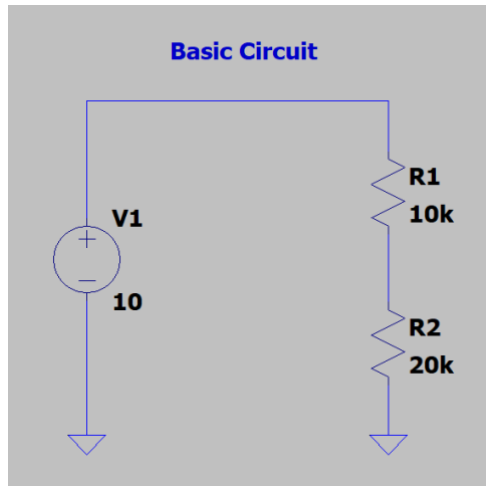
Applications include the study of rectifiers, amplifiers, filters, oscillators, and switching regulators.

PROCEDURE:

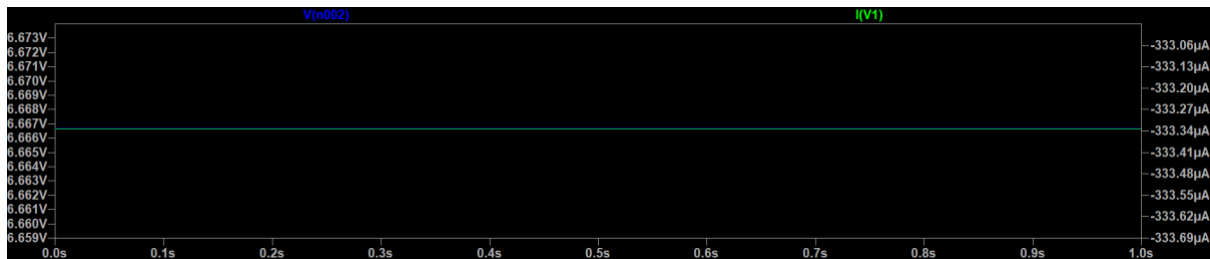
1. Install LTspice from the Analog Devices website.
2. Open the software → Create a New Schematic.
3. Select and place the required components from the component library (Resistors, Capacitors, Diodes, Transistors, Voltage Source, etc.).
4. Connect the components using the wire tool.
5. Set the component values by right-clicking on each element.
6. Define the type of analysis:
 - .op – DC operating point
 - .tran – Transient analysis
 - .ac – Frequency response analysis
7. Run the simulation by clicking the “Run” icon.

8. Observe waveforms of current/voltage in the Waveform Viewer.
9. Modify circuit parameters (e.g., resistance, supply voltage) and re-simulate to see the effect.

CIRCUIT:



SIMULATION:



RESULT: The simulation of the circuit in LTspice was successfully carried out, and the voltage and current waveforms were obtained.

PRECAUTIONS:

1. Ensure proper installation of LTspice.
2. Always check the polarity of sources and orientation of diodes/transistors before simulation.
3. Save the schematic frequently to avoid data loss.
4. Verify ground connection in every circuit as LTspice requires at least one ground node.

EXPERIMENT-2

AIM: To simulate and plot the I–V characteristics of a diode under forward bias and reverse bias conditions using LTspice.

APPARATUS: LTspice software

THEORY: A diode allows current to flow easily in one direction (forward bias) and blocks it in the opposite direction (reverse bias).

- **Forward bias:** When the anode is positive with respect to the cathode, the diode conducts after the threshold voltage (~ 0.7 V for silicon).
- **Reverse bias:** When the cathode is positive with respect to the anode, only a small leakage current flows until the reverse breakdown voltage is reached.

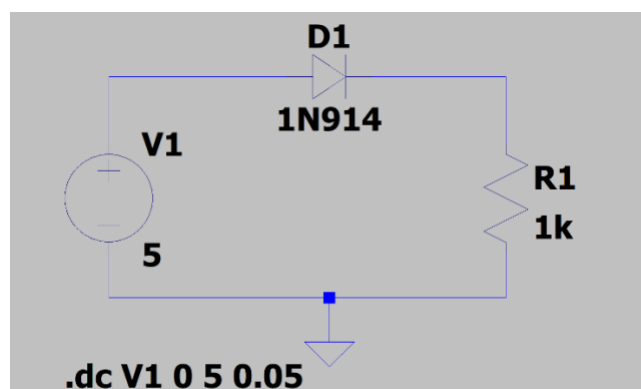
The I–V curve obtained by sweeping the voltage source shows the exponential rise in forward bias and nearly constant (small) leakage in reverse bias.

PROCEDURE:

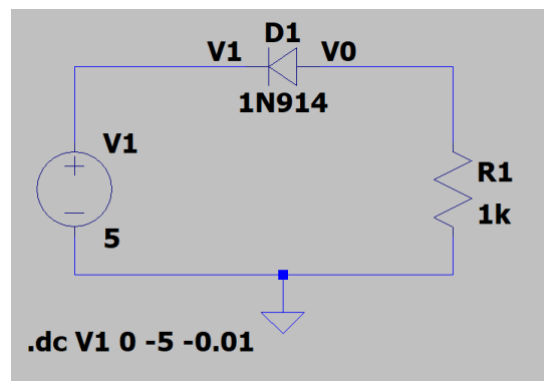
1. Open LTspice and draw the diode circuit with series resistor and DC voltage source.
2. Insert a .dc command to sweep the voltage across the diode.
3. For forward bias, sweep from **0 to +5 V**.
4. For reverse bias, sweep from **0 to –5 V**.
5. Run the simulation and plot diode current I(D1) versus diode voltage V(anode)–V(cathode).

CIRCUIT:

(a) Forward Bias

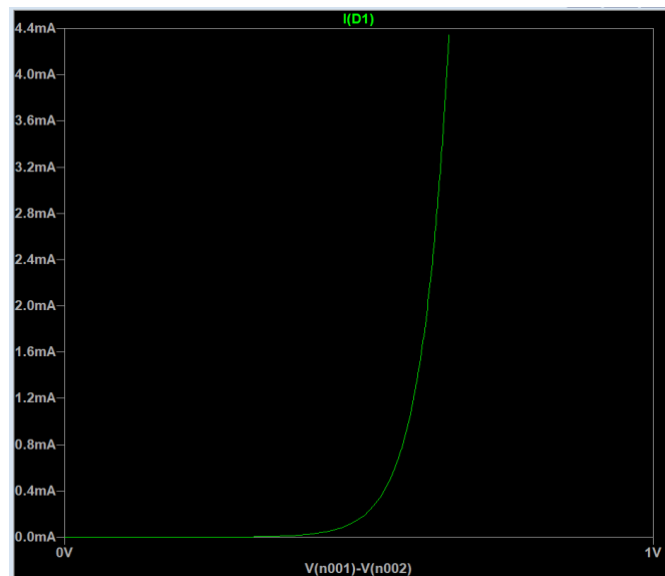


(b) Reverse Bias

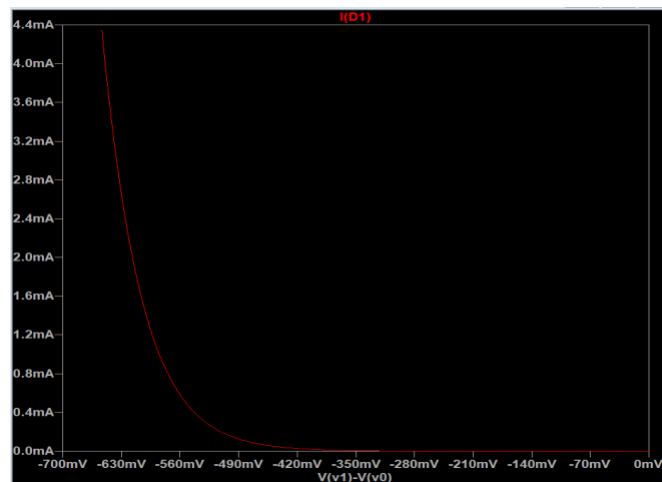


SIMULATION:

(a) Forward Bias



(b) Reverse Bias



RESULT: The simulated I–V characteristics of the diode match the expected theoretical behavior:

- In forward bias, the diode conducts significantly after ~ 0.7 V.
- In reverse bias, only a small leakage current flows until breakdown voltage is reached.

PRECAUTIONS:

1. The diode should be connected in the correct orientation for forward and reverse bias.
2. A series resistor must be used to limit current through the diode.
3. The applied voltage should not exceed the rated reverse breakdown voltage of the diode.

EXPERIMENT-3

AIM: To study the input characteristic (Base current I_b vs V_{BE}) of an NPN transistor in common-emitter configuration. To study the output characteristic (Collector current I_c vs V_{CE}) for different base drive (different V_{BE}) values.

COMPONENTS:

- NPN transistor BC547B (or similar)
- DC voltage source for base (V_{BE} sweep)
- DC voltage source for collector (V_{CE} sweep)
- Resistors: $R_1 = 1\text{ k}\Omega$ (base series), $R_2 = 1\text{ k}\Omega$ (collector series)
- LTspice software (simulation) (Values used in simulation: $R_1 = 1\text{ k}$, $R_2 = 1\text{ k}$, V_{BE} swept $0 \rightarrow 1\text{ V}$ step 0.1 V , V_{CE} swept $0 \rightarrow 5\text{ V}$ step 1 V)

THEORY:

In a common-emitter configuration the base-emitter junction behaves like a forward-biased diode. When V_{BE} is below $\sim 0.6\text{ V}$ the transistor is off (I_b and $I_c \approx 0$). Above $\sim 0.6\text{--}0.7\text{ V}$ base current flows and collector current starts increasing.

Input characteristic: plot of I_b vs V_{BE} (with fixed V_{CE}). It shows base diode threshold and base current rise.

Output characteristic: plot of I_c vs V_{CE} for different base drive levels (different V_{BE}). In active region I_c is approximately $\beta \times I_b$ and is fairly independent of V_{CE} (until saturation).

Current gain $\beta = I_c / I_b$ (measured in the active region, not in saturation).

PROCEDURE:

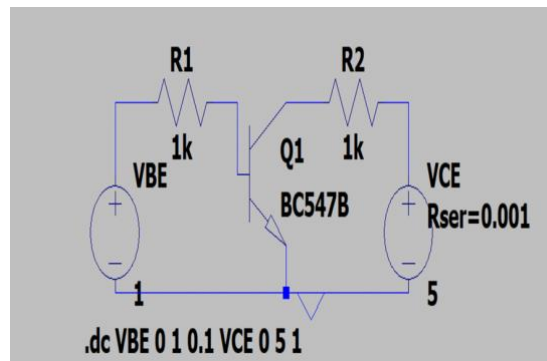
1. Build the CE circuit in LTspice with BC547B, $R_1 = 1\text{ k}\Omega$ in base and $R_2 = 1\text{ k}\Omega$ in collector. Emitter to ground.
2. Place voltage sources: V_1 connected to base node through R_1 (swept 0 to 1 V). V_2 connected to collector through R_2 (swept 0 to 5 V). Add the .dc directive: `.dc V1 0 1 0.1 V2 0 5 1`.
3. Run simulation.

In the plot window: Plot $I_b(Q1)$ vs V_1 (this gives base current vs V_{BE}). This is input characteristic (use a fixed V_{CE} or observe family of curves).

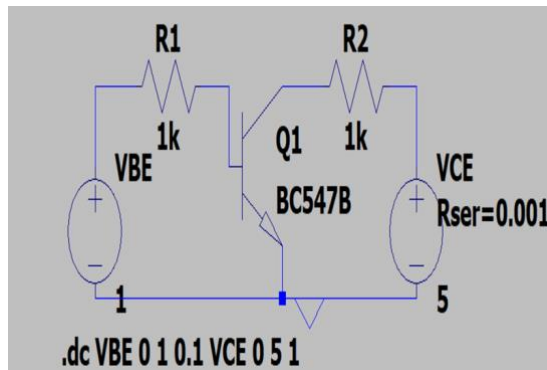
Plot $I(R_2)$ or $I_c(Q1)$ vs V_2 for different V_{BE} levels (each color trace corresponds to a different V_{BE}). This is output characteristic. Take values of I_b and I_c at chosen V_{BE} and V_{CE} and calculate $\beta = I_c / I_b$.

CIRCUIT:

(i)

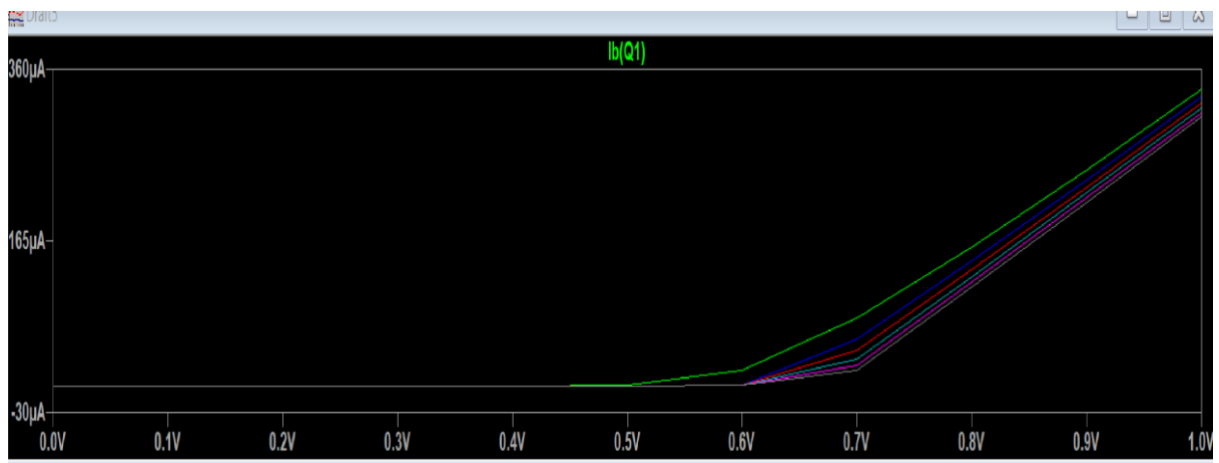


(ii)

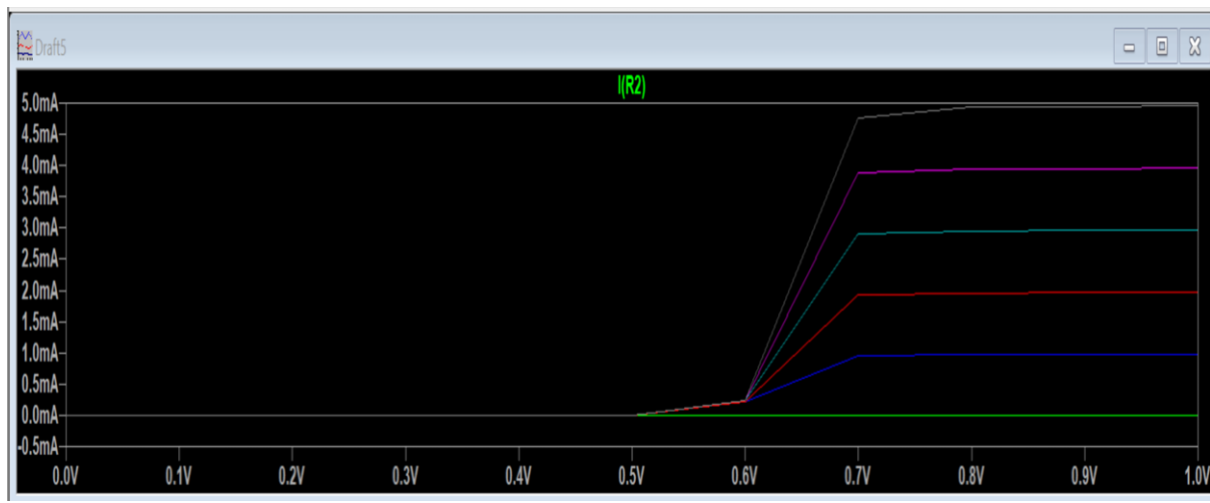


SIMULATION:

(i)



(ii)



OBSERVATIONS:

Input characteristic table

V_{BE} (V)	I_B (μ A) at $V_{CE}=2V$	I_B (μ A) at $V_{CE}=4V$	I_B (μ A) at $V_{CE}=6V$	I_B (μ A) at $V_{CE}=8V$	I_B (μ A) at $V_{CE}=10V$
0.1	0.0	0.0	0.0	0.0	0.0
0.2	0.0	0.0	0.0	0.0	0.0
0.3	0.0	0.0	0.0	0.0	0.0
0.4	0.0	0.0	0.0	0.0	0.0
0.5	1.0	1.0	1.0	1.0	1.0
0.6	20.0	19.0	18.0	17.0	16.0
0.7	60.0	58.0	56.0	55.0	53.0
0.8	120.0	118.0	115.0	113.0	110.0
0.9	200.0	198.0	195.0	193.0	190.0
1.0	320.0	318.0	315.0	313.0	310.0

Output characteristic table (I_C vs V_{CE} for several V_{BE} steps)

V_{CE} (V)	I_C (mA) at lowest base current	I_C (mA) at next base current	I_C (mA) at next base current	I_C (mA) at next base current	I_C (mA) at highest base current
0.5	0	0.2	0.5	0.8	1.0
0.6	0	0.4	0.8	1.2	1.5
0.7	0.1	0.8	1.5	2.5	3.5
0.8	0.3	1.0	2.0	3.0	4.0
0.9	0.3	1.0	2.0	3.0	4.5
1.0	0.3	1.0	2.0	3.0	5.0

RESULT: Base-emitter diode threshold $\sim 0.6\text{--}0.7$ V observed.

Collector current increases strongly when V_{BE} exceeds ~ 0.6 V. In active region (moderate V_{CE}), $I_c \approx \beta \times I_b$; β from simulation $\approx 20\text{--}200$ depending on operating point (report your measured values).

Saturation occurs if V_{CE} is too low (collector current becomes limited).

PRECAUTIONS:

1. Ensure emitter is at ground reference.
2. Use reasonable resistor values to avoid excessive currents that can damage a real transistor (in simulation $R_1, R_2 = 1k$ are fine).
3. In LTspice, name voltage sources uniquely (V_1, V_2). Do not use a single letter V as a numeric value. When sweeping two sources, use `.dc V1 start stop step V2 start stop step`.
4. Read currents with the correct node/component probe: $I_b(Q1)$ for base current, $I_c(Q1)$ or $I(R2)$ for collector current.

EXPERIMENT-4

AIM: To verify the truth tables of AND, OR, NAND, NOR, XOR, XNOR and NOT gates.

COMPONENTS:

- Breadboard
- DC power supply (5 V)
- Logic ICs (or discrete gate ICs) such as:
 1. 7408 (AND) or equivalent
 2. 7432 (OR) or equivalent
 3. 7400 (NAND) or equivalent
 4. 7402 (NOR) or equivalent
 5. 7486 (XOR) or equivalent
 6. 7485/other for XNOR (or use XOR + inverter to obtain XNOR)
 7. 7404 (NOT / Inverter)
- LED indicators (7–8) with series resistors ($\approx 330\ \Omega$)
- Connecting wires / jumpers
- Switches or jumpers to provide logic inputs (0/1)
- Multimeter (optional) to verify supply voltage

THEORY:

Digital logic gates perform basic Boolean operations on one or more input signals to produce an output signal. Each gate has a defined truth table that maps all possible input combinations to outputs. The basic gates are:

- **AND:** Output is HIGH only if *all* inputs are HIGH.
- **OR:** Output is HIGH if *any* input is HIGH.
- **NAND:** Complement (NOT) of AND. Output is LOW only if all inputs are HIGH.
- **NOR:** Complement of OR. Output is HIGH only when *no* inputs are HIGH.
- **XOR (Exclusive OR):** Output is HIGH when the number of HIGH inputs is odd (for 2-input XOR: HIGH when inputs differ).
- **XNOR (Exclusive NOR):** Complement of XOR. Output is HIGH when inputs are equal.
- **NOT (Inverter):** Produces the logical complement of the input.

PROCEDURE:

1. **Power setup:** Connect the ICs' VCC to +5 V and GND to 0 V. Verify power with a multimeter.
2. **Prepare inputs:** Use two SPDT switches (or jumper wires) for inputs A and B to toggle between logic 0 (GND) and logic 1 (+5 V). For the NOT gate, only one input is needed.
3. **Connect output indicators:** Connect an LED (with a 330 Ω resistor in series) between the gate output and GND (or to VCC for active-low outputs as appropriate). Confirm LEDs light correspondingly.
4. **Test all combinations:** For each gate, set inputs A and B to every possible combination (00, 01, 10, 11) and record the output. For the NOT gate, test input 0 and 1.
5. **Record observations:** Fill the observation table for each gate with the observed output (LED ON = logic 1; LED OFF = logic 0).
6. **Repeat / verify:** If any output does not match the expected truth table, double-check wiring, power connections, and IC pinouts.

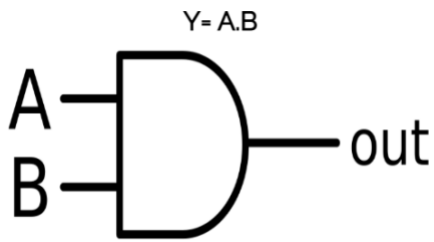
BOOLEAN EXPRESSIONS:

- AND ($A \cdot B$ or $A \wedge B$) = A AND B
- OR ($A + B$ or $A \vee B$) = A OR B
- NAND = $\neg(A \cdot B)$
- NOR = $\neg(A + B)$
- XOR = $A \oplus B = (A \cdot \neg B) + (\neg A \cdot B)$
- XNOR = $\neg(A \oplus B) = (A \cdot B) + (\neg A \cdot \neg B)$
- NOT = $\neg A$

TRUTH TABLES:

1. AND Gate

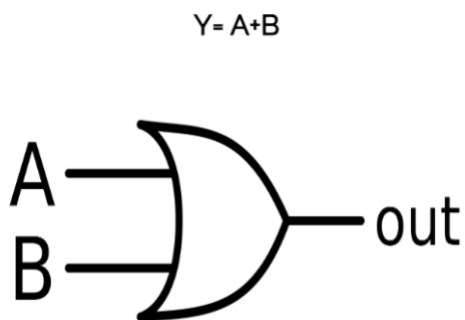
The AND gate is an electronic circuit that gives a high output (1) only if all its inputs are high. A dot (.) is used to show the AND operation i.e. $A \cdot B$ or can be written as AB .



A	B	$Y = A.B$
0	0	0
0	1	0
1	0	0
1	1	1

2. OR Gate

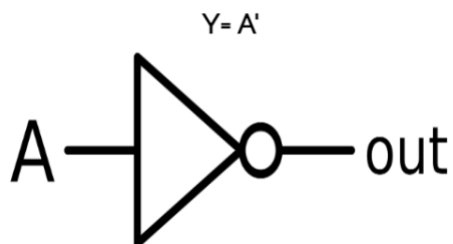
The OR gate is an electronic circuit that gives a high output (1) if one or more of its inputs are high. A plus (+) is used to show the OR operation.



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

3. NOT Gate

The NOT gate is an electronic circuit that produces an inverted version of the input at its output. It is also known as an inverter. If the input variable is A, the inverted output is known as NOT A. This is also shown as A' or A with a bar over the top, as shown at the outputs.

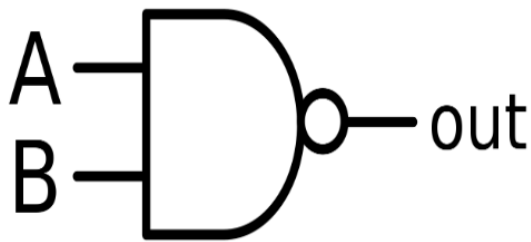


A	Y
0	1
1	0

4. NAND Gate

This is a NOT-AND gate which is equal to an AND gate followed by a NOT gate. The outputs of all NAND gates are high if any of the inputs are low. The symbol is an AND gate with a small circle on the output. The small circle represents inversion.

$$Y = \overline{AB}$$

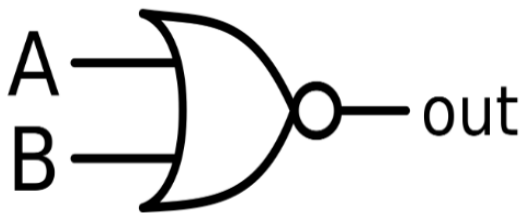


A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

5. NOR Gate

This is a NOT-OR gate which is equal to an OR gate followed by a NOT gate. The outputs of all NOR gates are low if any of the inputs are high. The symbol is an OR gate with a small circle on the output. The small circle represents inversion.

$$Y = \overline{A+B}$$

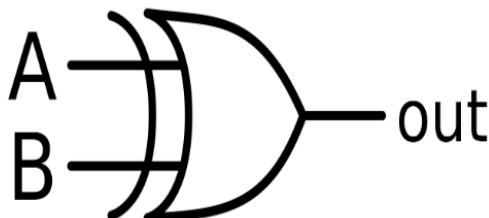


A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

6. XOR Gate

The 'Exclusive-OR' gate is a circuit which will give a high output if either, but not both of its two inputs are high. An encircled plus sign ($\oplus$) is used to show the Ex-OR operation.

$$Y = A \oplus B$$

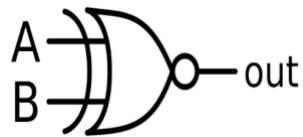


A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

7. XNOR Gate

The 'Exclusive-NOR' gate circuit does the opposite to the EX-OR gate. It will give a low output if either, but not both of its two inputs are high. The symbol is an EX-OR gate with a small circle on the output. The small circle represents inversion.

$$Y = \overline{A \oplus B}$$



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

RESULT: The experimentally observed outputs matched the standard truth tables for the AND, OR, NAND, NOR, XOR, XNOR, and NOT gates when the circuit wiring and power were correct.

PRECAUTIONS:

1. Double-check IC pinouts before connecting.
2. Ensure VCC and GND are connected correctly to the ICs.
3. Use correct series resistors with LEDs to prevent damage.
4. Avoid short-circuiting power rails.

EXPERIMENT-5

AIM: To design AND, OR, NOT, XOR, XNOR, NAND and NOR gates using only NAND gates.

COMPONENTS: IC 7400 (NAND gate IC), connecting wires and bread board.

THEORY:

The NAND gate is known as a universal gate, meaning any logic function can be implemented using only NAND gates.

A NAND gate outputs LOW only when all inputs are HIGH.

Its Boolean expression is:

$$Y = \bar{A}\bar{B}$$

1. OR GATE

OR gate can be designed using 3 NAND gates. For OR gate with A and B as input and Y the output then

$$Y = A + B$$

$$Y = ((A + B)')' = (A' \cdot B')' \text{ (using Demorgan's Law)}$$
$$= (A \text{ NAND } A) \text{ NAND } (B \text{ NAND } B)$$

2. AND GATE

AND gate can be designed using 2 NAND gates.

$$\text{Here } Y = A \cdot B$$

$$Y = ((A \cdot B)')' = (A \text{ NAND } B) \text{ NAND } (A \text{ NAND } B)$$

3. NOT GATE

NOT gate can be designed using only 1 NAND gate. For NOT gate:

$$Y = A' \quad Y = ((A)')' = (A \cdot A)'$$

$$Y = A \text{ NAND } A$$

4. NOR GATE

NOR gate can be designed using 4 NAND gates. For NOR gate:

$$Y = (A + B)'$$

$$Y = [(A \text{ NAND } A) \text{ NAND } (B \text{ NAND } B)] \text{ NAND } [(A \text{ NAND } A) \text{ NAND } (B \text{ NAND } B)]$$

5.XOR GATE

XOR gate can be designed using 4 NAND gates. For XOR gates:

$$Y = A'B + AB'$$

$$Y = [(A \text{ NAND } (A \text{ NAND } B)) \text{ NAND } (B \text{ NAND } (A \text{ NAND } B))]$$

6. XNOR GATE

XNOR gate can be designed using 5 NAND gates.

$$Y = [(A \text{ NAND } (A \text{ NAND } B)) \text{ NAND } (B \text{ NAND } (A \text{ NAND } B))] \text{ NAND } [(A \text{ NAND } (A \text{ NAND } B)) \text{ NAND } (B \text{ NAND } (A \text{ NAND } B))]$$

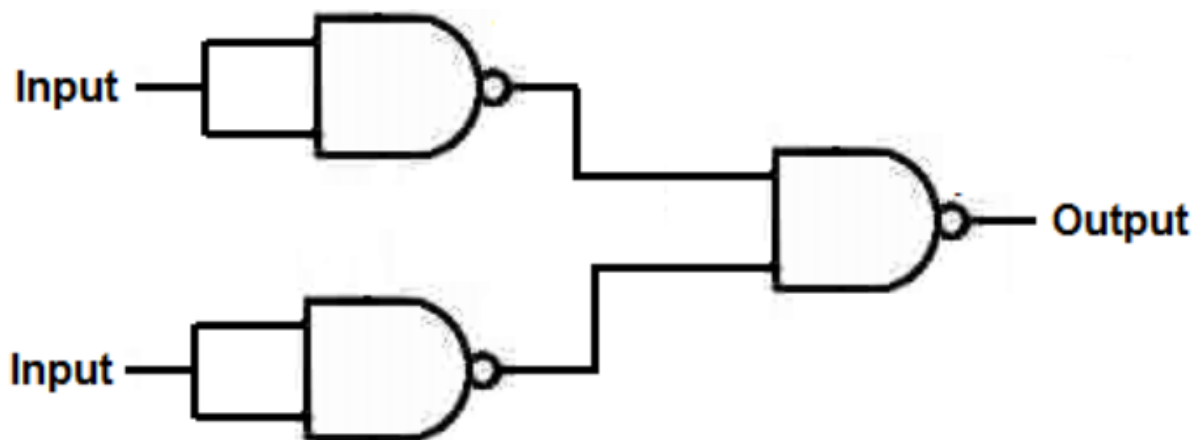
PROCEDURE:

1. For designing the required gate from NAND gate, connect the required IC.
2. Connect the gate in the required manner with the help of connecting wires and make proper connections for ground (pin 7) and supply (pin 14).
3. Connect the inputs and outputs to the required LEDs. Switch on the supply.
4. Note the output for various combinations of input, note the observations and compare them with the observations of the original gate.
5. Repeat the above steps for all other gates and note the observations.

OBSERVATIONS:

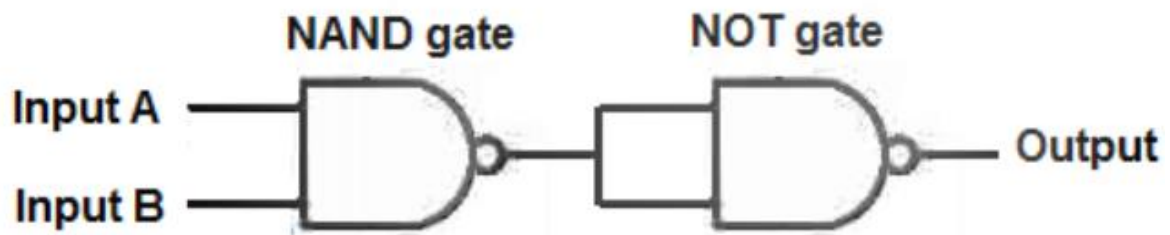
(1) OR GATE

$$Y = A + B$$



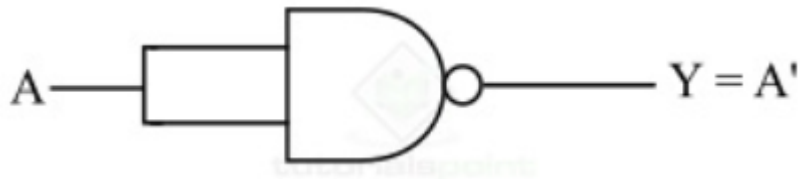
(2) AND GATE

$$Y = A.B$$



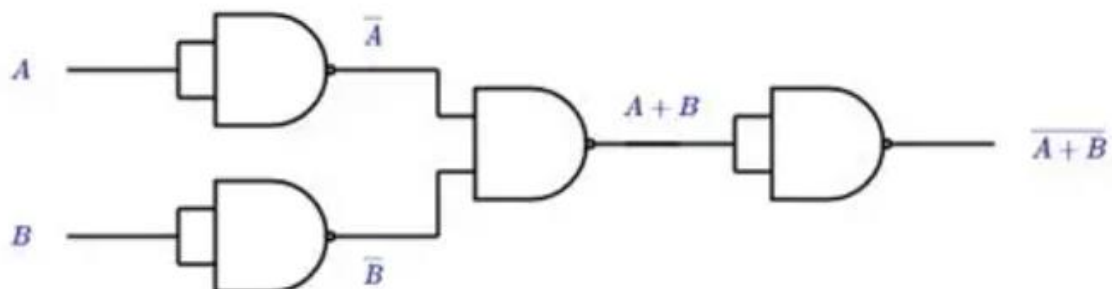
(3) NOT GATE

$$Y = A'$$



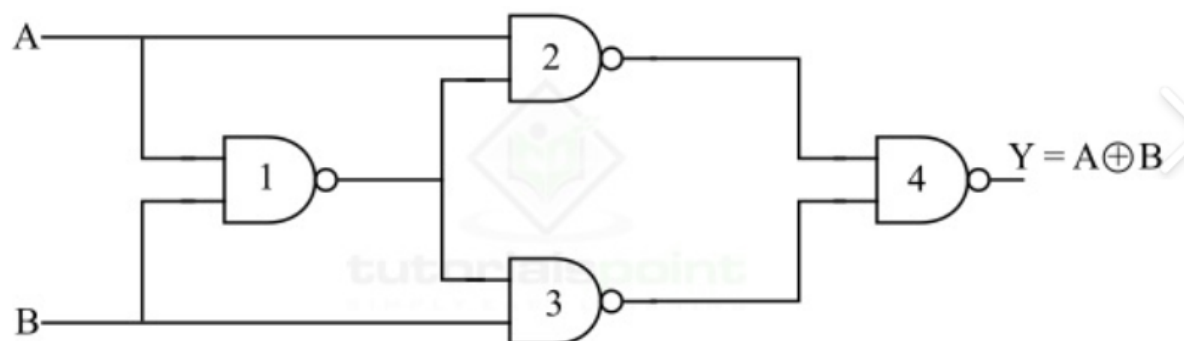
(4) NOR GATE

$$Y = (A+B)'$$



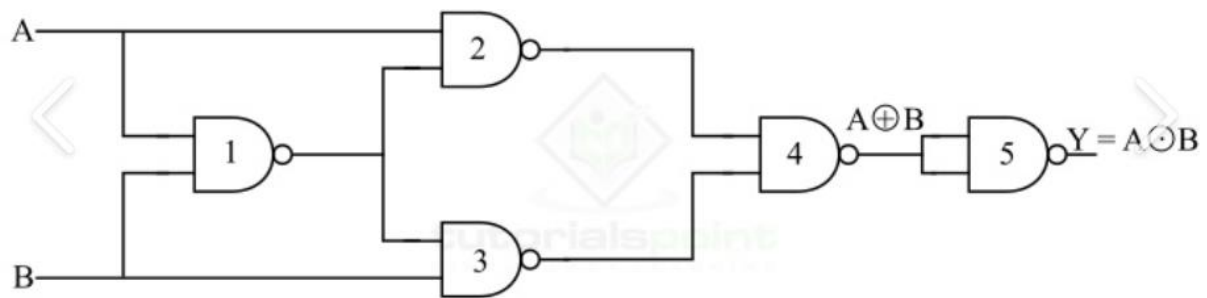
(4) XOR GATE

$$Y = A'B + AB'$$



(5) XNOR GATE

$$Y = (A'B + AB')'$$



RESULT: All the gates (AND, OR, NOT, XOR, XNOR, NOR) were designed using NOR gates and their truth tables verified.

PRECAUTIONS:

1. All the IC's should be handled carefully.
2. All the connections should be tight.
3. Supply should be given after all connections are made.
4. Use IC plucker when removing the IC from the breadboard.

EXPERIMENT-6

AIM: To design AND, OR, NOT, XOR, XOR, NAND (NOR) gates using NOR (IC No-7402) gate.

COMPONENTS: IC 7402 (NOR gate IC), connecting wires and bread board.

THEORY: It is possible to design all logic gates using NOR gates that is why it is called a universal gate.

1. OR Gate

OR GATE OR gate can be designed using 2 NOR gates. Here $Y = (A \text{ NOR } B) \text{ NOR } (A \text{ NOR } B)$.

2. AND GATE

AND gate can be designed using 3 NOR gates. Here $Y = (A \text{ NOR } A) \text{ NOR } (B \text{ NOR } B)$.

3. NOT GATE

NOT gate can be designed using only 1 NOR gate. For NOT gate: $Y = A \text{ NOR } A$

4. NAND GATE

NAND gate can be designed using 4 NOR gates. $Y = [(A \text{ NOR } A) \text{ NOR } (B \text{ NOR } B)] \text{ NOR } [(A \text{ NOR } A) \text{ NOR } (B \text{ NOR } B)]$

5. XOR GATE

XOR gate can be designed using 5 NOR gates. For XOR gates: $Y = [((A \text{ NOR } (A \text{ NOR } B)) \text{ NOR } (B \text{ NOR } (A \text{ NOR } B))) \text{ NOR } ((A \text{ NOR } (A \text{ NOR } B)) \text{ NOR } (B \text{ NOR } (A \text{ NOR } B)))]$

6. XNOR GATE

XNOR gate can be designed using 4 NOR gates. Here

$Y = [(A \text{ NOR } (A \text{ NOR } B)) \text{ NOR } (B \text{ NOR } (A \text{ NOR } B))]$

PROCEDURE:

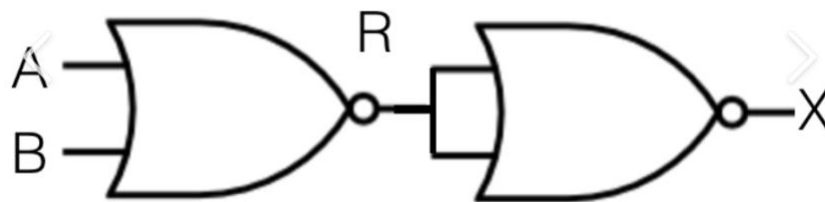
1. For designing the required gate from NOR gate, connect the required IC.
2. Connect the gate in the required manner with the help of connecting wires and make proper connections for ground (pin 7) and supply (pin 14).

3. Connect the inputs and outputs to the required LEDs. Switch on the supply.
4. Note the output for various combinations of input, note the observations and compare them with the observations of the original gate.
5. Repeat the above steps for all other gates and note the observations.

OBSERVATIONS:

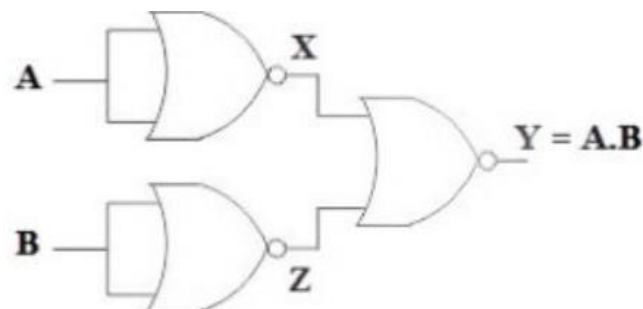
(1) OR GATE

$$Y = A + B$$



(2) AND GATE

$$Y = A.B$$



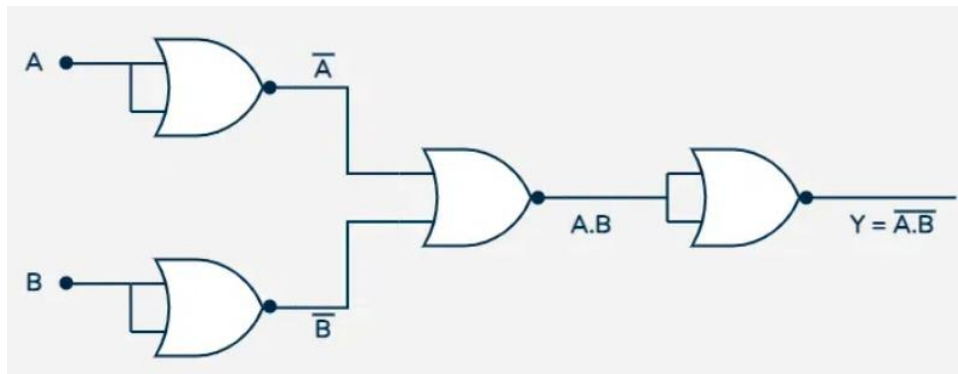
(3) NOT GATE

$$Y = A'$$



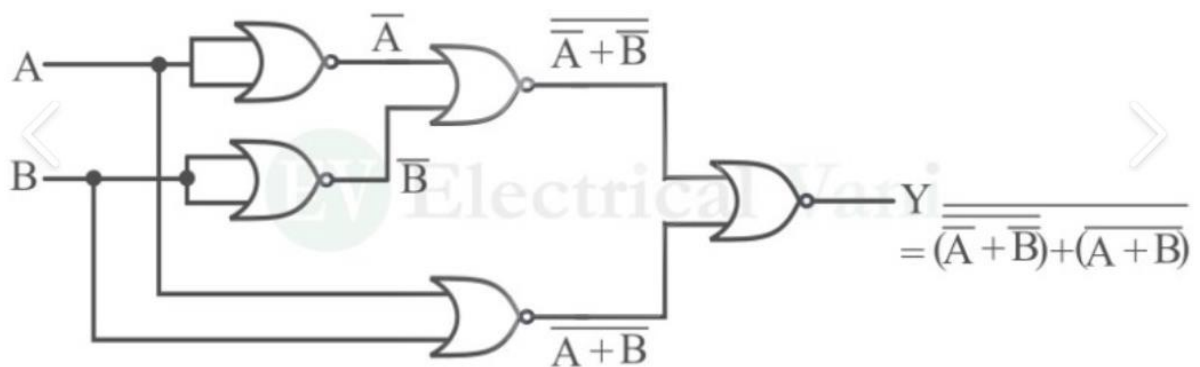
(4) NAND GATE

$$Y = (A.B)'$$



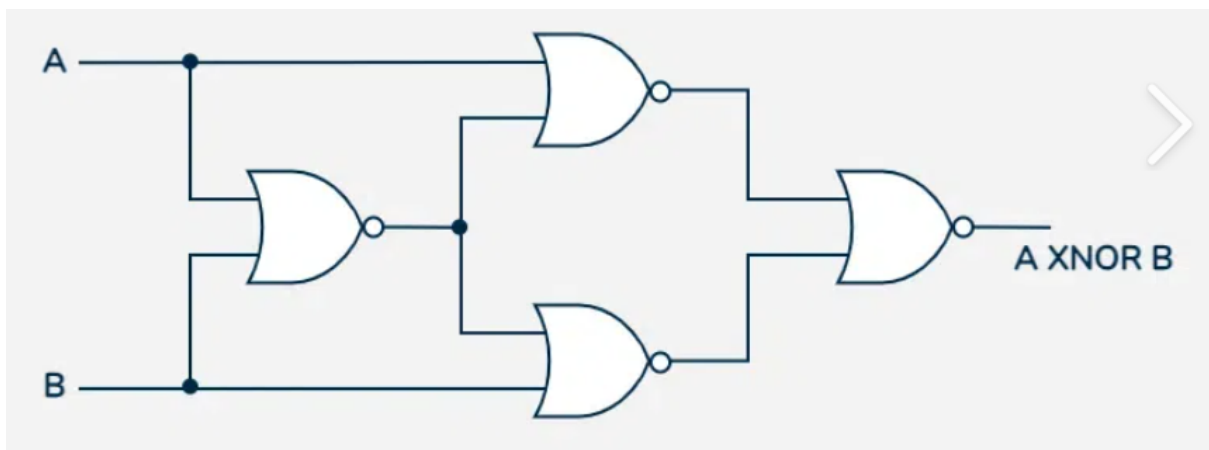
(5) XOR GATE

$$Y = A'B + AB'$$



(6) XNOR GATE

$$Y = (A'B + AB')'$$



RESULT: All the gates (AND, OR, NOT, XOR, XNOR, NAND) were designed using NOR gates and their truth tables verified.

PRECAUTIONS:

1. All the IC's should be handled carefully.
2. All the connections should be tight.
3. Supply should be given after all connections are made.
4. Use IC plucker when removing the IC from the breadboard.

EXPERIMENT-7

AIM: To design a half adder circuit and verify its truth table.

COMPONENTS: IC No. 7432(OR Gate), 7408(AND Gate), and 7486 (XOR Gate), connecting wires and trainers kit.

THEORY: A half adder is a combinational circuit that programs the addition of two bits. The circuit has two binary inputs and two binary outputs. The input variables (A and B) designate the augend and addended bits and the output variables (S and C) are sum and carry respectively. The C output is 1 only when both are 1 and S represents the last significant bit of the sum.

The simplified Boolean functions for two outputs as obtained from the truth table are:

$$S=A'B+AB' \quad C=AB$$

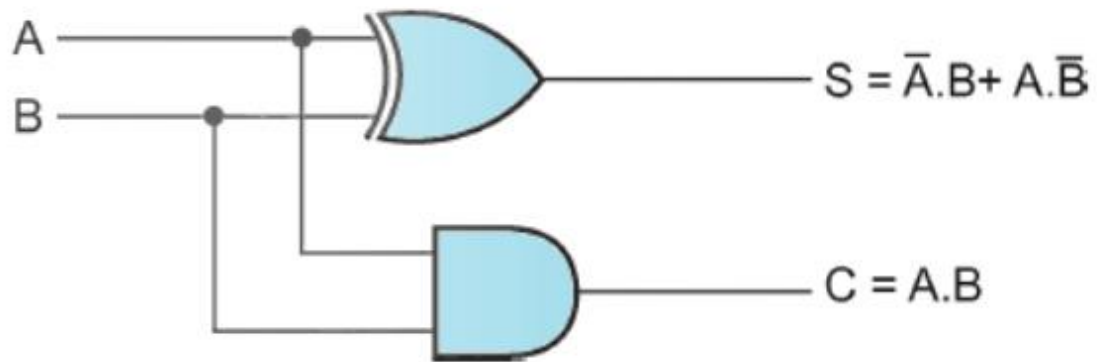
PROCEDURE:

1. For designing half adder, connect the required IC.
2. Connect the gates in the required manner with the help of connecting wires and make proper connections for ground (pin 7) and supply (pin 14).
3. Connect the input and output to required LEDs. Switch on the supply.
4. Note the output for various combinations of input variables.

OBSERVATIONS:

Truth Table of Half Adder:

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



RESULT: A half adder was designed and its truth table was verified. A full adder was also designed using half adders and its truth table is verified.

PRECAUTIONS:

1. All the IC's should be handled carefully.
2. All the connection should be tight.
3. Supply should be given after all connections are made.
4. Use IC plucker when remove the IC from Bread Board.

EXPERIMENT-8

AIM: To design a half subtractor circuit and verify its truth table.

COMPONENTS: IC No. 7432(OR Gate), 7408(AND Gate), and 7486 (XOR Gate), 7404 (NOT Gate) connecting wires and trainers kit.

THEORY: A half subtractor is a combinational circuit that programs the subtraction of two bits. The circuit has two binary inputs (A and B) and two binary outputs(D as difference and B0 as borrow).

The simplified Boolean functions for two outputs as obtained from the truth table are:

$$D = A'B + AB'$$

$$B0 = A'B$$

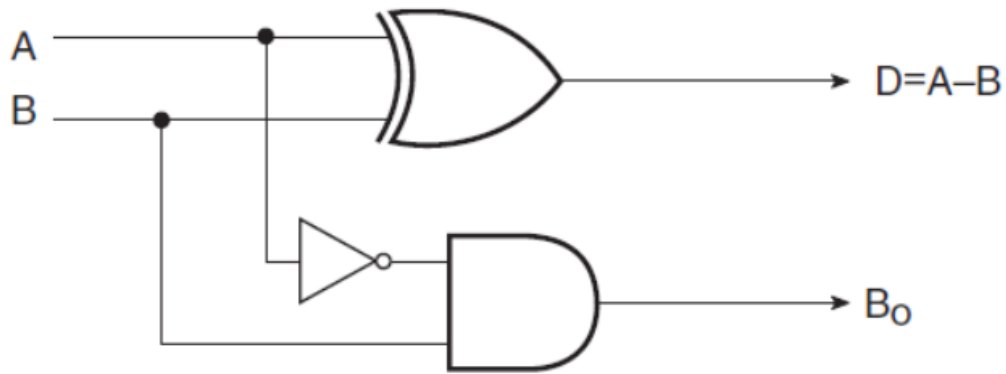
PROCEDURE:

- 1.Connect the required ICs and make connections of gates in required manner and make proper connections for ground (pin 7) and supply (pin 14).
- 2.Connect the input and output to required LEDs. Switch on the supply.
- 3.Note the output for various combinations of input variables.

OBSERVATIONS:

Truth Table of Half Subtractor:

A	B	D	B0
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0



RESULT: A half subtractor was designed and its truth table was verified.

PRECAUTIONS:

5. All the IC's should be handled carefully.
6. All the connection should be tight.
7. Supply should be given after all connections are made.
8. Use IC plucker when remove the IC from Bread Board.